Inverse Mechanical Mathematics

The Axiom of 1, the Parity Rule, and the Count-by-Count Unfolding of the Nine Active Depths

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Axioms of the Inverse Mechanical Chain

Axiom I (The Axiom of 1). There is exactly one object: the unit 1. It has several equivalent views,

$$1 = -1 = \sqrt{1} = \sqrt{2^0},$$

each a different face of the same single whole. All structure in the chain is unfolded from this single object by inward partition.

Axiom II (Halving and Parity). The only operation is halving. Each halving generates *two* structures, and each structure carries *two* readings under operational counting: one rational and one irrational. The reading at any local depth $\sqrt{2^d}$ is rational when the exponent d is even and irrational (carrying a factor of $\sqrt{2}$) when d is odd.

Halving generates the structures. Parity determines the readings.
Together they unfold an inverse chain of operational counts.

The Only Two Letters: c and d

This framework uses only c (the count) and d (the depth). Standard placeholder indices like k, i, or n are not used, because they carry no mechanics — c and d already do. c is the exact halving step, and d is the exact local depth whose parity forces the reading. The labels *are* the mechanics.

This paper works the **full first closure**: five active counts $c \in \{1, \ldots, 5\}$ unfolding nine active depths $d \in \{1, \ldots, 9\}$, seated alongside the axiom at rest (labelled $c = 0$, $d = 0$, the unit $1 = \sqrt{2^0}$ with no partition performed). Both c and d continue past this closure in principle — halving keeps going, and each new count unfolds deeper depths. Higher counts, deeper depths, and their mechanics are reserved for future papers.

What This Construction Is, and What It Is Not

Several existing frameworks look superficially similar to the inverse mechanical chain and are worth distinguishing at the outset, so that readers familiar with them do not pattern-match incorrectly.

Not 2-adic analysis. The 2-adic numbers have a nested scale structure, but they are built the opposite way: starting from the integers and defining closeness by divisibility. The 2-adic construction does not unfold from 1 as a whole, and it does not generate depth-within-depth as a mechanical consequence of a count. It imports its structure from outside. The inverse mechanical chain does not.

Not measure theory. Measure theory partitions a whole, but its partitions are chosen — a sigma-algebra is selected by the mathematician, and the whole is then decomposed against that choice. The inverse mechanical chain has no such choice. The partitions are not selected; they are what the count does at each step. The count is the partition.

Not dyadic expansion. The dyadic expansion of $[0, 1]$ is a flat binary addressing of the unit interval. It halves repeatedly, but it does not carry the structure of the count that produced it. There is no notion in the dyadic expansion of "count c owns depth d," no parity rule forcing readings, and no conservation across domains. The expansion is an address; the inverse mechanical chain is a mechanism.

What the chain does. The count generates the depths. Each depth carries its own substructure by the same rule that produced the shallower depths. The parity rule is not imposed from outside — it is what the count does at each step. Every depth mechanically contains the structure of every shallower depth, because the same operation produced them all. The layering continues as long as the count continues, and closes on a conservation law at every level. The whole paper is a single object — the axiom of 1 — read at different depths of its own unfolding. Every part that follows (the singular and dual counts, the terminal pair, the four parallel faces, the machined integer 720, the constant per-unit slip) is the same 1 seen under the same rule, deeper in.

The chain shares ingredients with frameworks that partition, halve, or nest. It is not any of them. It is a count-driven, depth-generating, parity-organized unfolding of a single whole, and it is the object this paper describes.

Abstract

We begin at the axiom of 1, and nothing else. The single object is $1 = \sqrt{2^0}$, the unit at rest. From this starting point the inverse mechanical count unfolds by partition inward. Each active step of the count lays down a new local depth on the exponent ladder $\sqrt{2^1}, \sqrt{2^2}, \ldots, \sqrt{2^9}$, and the reading at each local depth is forced by one rule:

Parity rule: at local depth $\sqrt{2^0}, \sqrt{2^2}, \sqrt{2^4}, \sqrt{2^6}, \sqrt{2^8}$ the reading is *rational*; at local depth $\sqrt{2^1}, \sqrt{2^3}, \sqrt{2^5}, \sqrt{2^7}, \sqrt{2^9}$ the reading is *irrational* (carries a factor of $\sqrt{2}$).

Even exponents give rational readings; odd exponents give irrational readings. This is the only causal rule in the chain. Everything else — singular counts, dual counts, the terminal pair, the domain multiplicity, the machining of 720, and the outside mirror chain — is a consequence of the parity rule applied to the axiom of 1. The count unfolds as follows. The axiom sits at rest at $c = 0$, $d = 0$: exponent 0 is even, reading is the unit 1, and no partition has been performed. Count 1 is the first active halving and owns the single local depth $\sqrt{2^1} = \sqrt{2}$. The exponent 1 is odd, so count 1 is an irrational singular: one reading, value $\sqrt{2}$, two domains. From count 2 onward, each count owns two consecutive local depths. Because consecutive whole-number exponents always contain one even exponent and one odd exponent, every count from 2 through 5 is forced to carry one rational reading and one irrational reading *simultaneously* — a dual count. The dual nature is not a separate axiom; it is the direct consequence of owning two

consecutive depths under the parity rule. The count terminates at count 5, which owns local depths $\sqrt{2^8}$ and $\sqrt{2^9}$. At that point the nine active depths $d = 1$ through 9 are fully unfolded, seated alongside the axiom at rest, and the chain has produced the terminal pair $(16,\ 16\sqrt{2})$. The active-depth tally is therefore

$$\underbrace{1}_{\substack{\text{singular}\\ c=1}} + \underbrace{2}_{\substack{\text{dual}\\ c=2}} + \underbrace{2}_{\substack{\text{dual}\\ c=3}} + \underbrace{2}_{\substack{\text{dual}\\ c=4}} + \underbrace{2}_{\substack{\text{dual}\\ c=5}} = 9 \text{ active depths,}$$

produced by the five active counts ($c = 1$ through 5) and seated alongside the axiom at rest ($c = 0$, $d = 0$ — no partition, no action). All mechanical work is performed by the five halvings unfolding the nine active depths. This paper develops the mechanics of the count-by-count unfolding across seven parts. Parts One through Six trace the inverse mechanical count from the axiom of 1 to the closed mirror engine at the shared terminal pair $(16, 16\sqrt{2})$, while Part Seven interprets the finished structure as a 4D inverse tensor in the framework of Inverse Dual Hemisphere Quantum Mechanics. In this unfolding the entire structure is revealed to be perfectly reflective: every element mirrors the parity rule applied to the original axiom, and the closed engine reflects itself inward and outward in complete self-consistency.

Notation

The only letters used in this framework are c and d. Every other symbol in the chain is either a pure number or the fixed mark $\sqrt{\cdot}$. There are no invented variables, no named constants, and no auxiliary parameters. The two letters stand for:

- c — the **count**, a partition step (halving) of the axiom. Active counts take the whole-number values $1, 2, 3, 4, 5$. The label $c = 0$ is reserved for the axiom seat (no partition performed).
- d — the **depth**, a local depth on the exponent ladder $\sqrt{2^d}$. Active depths take the whole-number values $1, 2, 3, 4, 5, 6, 7, 8, 9$. The label $d = 0$ is reserved for the axiom seat (the unit at rest, $\sqrt{2^0} = 1$).

Everything else in the paper is a direct reading produced by the two axioms: the axiom of 1 and the halving/parity rule. With that fixed, the following shorthand is used throughout:

1	the axiom; the single object $1 = \sqrt{2^0}$.
$\sqrt{2^d}$	local depth d. Active exponents take the whole-number values $1, 2, 3, 4, 5, 6, 7, 8, 9$.
count	a partition step (halving) of the axiom. Active counts take the whole-number values $1, 2, 3, 4, 5$.
count c owns depth d	count c is the count that unfolds local depth d.
parity rule	at even exponents the reading is rational; at odd exponents the reading is irrational.
rational reading	a reading with no $\sqrt{2}$ factor; plain whole or rational number.
irrational reading	a reading carrying one factor of $\sqrt{2}$.
singular count	a count that owns one local depth, therefore carries one reading.
dual count	a count that owns two consecutive local depths, therefore carries one rational reading and one irrational reading at once.
domain	one of the pieces the axiom has been partitioned into at the current count.

Active depths are unfolded in strict whole-number order $1, 2, 3, 4, 5, 6, 7, 8, 9$, starting from the axiom at rest. Active counts unfold in strict whole-number order $1, 2, 3, 4, 5$, also starting from the axiom at rest. The number of domains at count c is exactly 2^c, taking the whole-number values $1, 2, 4, 8, 16, 32$ as the count advances from the axiom seat through the five active halvings.

Part One — The Count-by-Count Unfolding

This part walks the chain one count at a time. Each active count unfolds one or two local depths. For each depth we state the exponent, the parity (even or odd), the reading forced by the parity rule, the value, and the number of domains active at that count.

The Axiom Seat and the Active Chain

Before walking the chain, one distinction must be fixed once and for all. The labels $c = 0$ and $d = 0$ are *not* a count and a depth produced by the chain — they are the axiom itself. No partition has been performed at $c = 0$, and no new depth has been produced at $d = 0$. The axiom is the unit at rest; it is the starting condition, not mechanical work.

The mechanical work of the chain is therefore done by:

- **Five active counts**: $c = 1, 2, 3, 4, 5$ — the five halvings.
- **Nine active depths**: $d = 1, 2, 3, 4, 5, 6, 7, 8, 9$ — the nine new depths produced by those halvings.

This gives the active product $5 \times 9 = 45$, the base mechanical unit that will machine the integer 720 in Part Four.

The axiom seat sits alongside the active chain as the unit at rest. When this paper labels $c = 0$ or $d = 0$ in a table, it labels the axiom seat — not a count or a depth that the chain has produced. All mechanical statements in the paper refer to the five active halvings and the nine active depths.

1 Count 0 — The Axiom at Rest

The axiom is the unit at rest. No partition has been performed, so there is exactly one domain: the whole of 1.

- Axiom seat labelled $c = 0$, $d = 0$, $\sqrt{2^0} = 1$.
- Exponent 0 is even; reading is the rational unit 1.
- Domains: 1.

The axiom seat is a **rational singular** reading — one unit, one domain — but it is not an active count or an active depth. It is the axiom of 1 read on the exponent ladder, seated underneath everything the chain will produce.

Label c	**Label** d	$\sqrt{2^d}$	**Value**	**Parity** $\rightarrow$ **Reading**
0	0	$\sqrt{2^0}$	1	even $\rightarrow$ rational

Table 1: The axiom seat at $c = 0$, $d = 0$. Unit at rest, no partition.

2 Count 1 — The First Active Halving at Depth 1

Count 1 performs the first partition of the axiom. The whole of 1 is halved into two domains. Count 1 owns the first active local depth on the exponent ladder: $\sqrt{2^1}$.

- Count 1 owns local depth $\sqrt{2^1}$.
- Exponent 1 is odd, so by the parity rule the reading is **irrational**.
- Reading: $\sqrt{2} \approx 1.414214$.
- Domains: 2.

Count 1 is an **irrational singular**: one depth, one reading, two domains. This is the first place on the ladder where $\sqrt{2}$ appears, and it appears because exponent 1 is the first odd exponent.

Count c	**Depth** d	$\sqrt{2^d}$	**Value**	**Parity** $\rightarrow$ **Reading**
1	1	$\sqrt{2^1}$	$\sqrt{2} \approx 1.414214$	odd $\rightarrow$ irrational

Table 2: Count 1 unfolds local depth $d = 1$. Irrational singular.

3 Count 2 — The First Dual, Depths 2 and 3

Count 2 partitions again. From this count forward, every count owns two consecutive local depths, because the axiom keeps being halved and the exponent ladder keeps advancing by two whole numbers per partition.

Count 2 owns local depths $\sqrt{2^2}$ and $\sqrt{2^3}$. One of those exponents is even (2) and the other is odd (3), so by the parity rule count 2 carries *both* a rational reading *and* an irrational reading at the same count step. This is the structural definition of a dual count: it is forced by owning one even and one odd local depth at the same time.

- Count 2, depth $\sqrt{2^2}$, exponent 2 (even) — rational reading 2.
- Count 2, depth $\sqrt{2^3}$, exponent 3 (odd) — irrational reading $2\sqrt{2}$.
- Domains: 4.

Count c	**Depth** d	$\sqrt{2^d}$	**Value**	**Parity** $\rightarrow$ **Reading**
2	2	$\sqrt{2^2}$	2	even $\rightarrow$ rational
2	3	$\sqrt{2^3}$	$2\sqrt{2} \approx 2.828427$	odd $\rightarrow$ irrational

Table 3: Count 2 unfolds local depths $d = 2$ and $d = 3$. First dual count.

4 Count 3 — Dual, Depths 4 and 5

Count 3 partitions again. It owns local depths $\sqrt{2^4}$ and $\sqrt{2^5}$ — again one even exponent and one odd exponent.

- Count 3, depth $\sqrt{2^4}$, exponent 4 (even) — rational reading 4.
- Count 3, depth $\sqrt{2^5}$, exponent 5 (odd) — irrational reading $4\sqrt{2}$.
- Domains: 8.

Count c	**Depth** d	$\sqrt{2^d}$	**Value**	**Parity** $\rightarrow$ **Reading**
3	4	$\sqrt{2^4}$	4	even $\rightarrow$ rational
3	5	$\sqrt{2^5}$	$4\sqrt{2} \approx 5.656854$	odd $\rightarrow$ irrational

Table 4: Count 3 unfolds local depths $d = 4$ and $d = 5$.

5 Count 4 — Dual, Depths 6 and 7

Count 4 partitions again and owns local depths $\sqrt{2^6}$ and $\sqrt{2^7}$.

- Count 4, depth $\sqrt{2^6}$, exponent 6 (even) — rational reading 8.

- Count 4, depth $\sqrt{2^7}$, exponent 7 (odd) — irrational reading $8\sqrt{2}$.
- Domains: 16.

Count c	**Depth** d	$\sqrt{2^d}$	**Value**	**Parity** $\rightarrow$ **Reading**
4	6	$\sqrt{2^6}$	8	even $\rightarrow$ rational
4	7	$\sqrt{2^7}$	$8\sqrt{2} \approx 11.313708$	odd $\rightarrow$ irrational

Table 5: Count 4 unfolds local depths $d = 6$ and $d = 7$.

6 Count 5 — Dual, Terminal Depths 8 and 9

Count 5 is the terminal count. It partitions one last time and owns local depths $\sqrt{2^8}$ and $\sqrt{2^9}$. When count 5 unfolds the odd local depth $\sqrt{2^9}$, the full active exponent ladder 1 through 9 has been laid down and the chain is complete.

- Count 5, depth $\sqrt{2^8}$, exponent 8 (even) — rational reading 16.
- Count 5, depth $\sqrt{2^9}$, exponent 9 (odd) — irrational reading $16\sqrt{2}$.
- Domains: 32.

The terminal pair of the chain is $(16, 16\sqrt{2})$. The terminal exponent is 9. The five active halvings have unfolded all nine active depths; the axiom at $d = 0$ remains at rest beneath them.

Count c	**Depth** d	$\sqrt{2^d}$	**Value**	**Parity** $\rightarrow$ **Reading**
5	8	$\sqrt{2^8}$	16	even $\rightarrow$ rational
5	9	$\sqrt{2^9}$	$16\sqrt{2} \approx 22.627417$	odd $\rightarrow$ irrational

Table 6: Count 5 unfolds the terminal local depths $d = 8$ and $d = 9$. Terminal pair: $(16, 16\sqrt{2})$.

7 Parity Summary of the Chain

Every local depth has now been unfolded by the count that owns it. The parity rule forces the reading at each depth:

Count 1 is singular because it owns a single local depth at the odd exponent 1 and is therefore irrational. Counts 2, 3, 4, and 5 are dual because each one owns two consecutive local depths. Two consecutive whole-number exponents always contain one even and one odd exponent, so every dual count is forced to carry one rational reading and one irrational reading at once. No extra axiom is introduced; the structure of the chain is entirely a consequence of the parity rule applied to the axiom of 1.

Count c	**Depth** d	$\sqrt{2^d}$	2^d	**Parity** $\rightarrow$ **Reading**	**Value**
0	0	$\sqrt{2^0}$	1	even $\rightarrow$ rational (axiom)	1
1	1	$\sqrt{2^1}$	2	odd $\rightarrow$ irrational	$\sqrt{2}$
2	2	$\sqrt{2^2}$	4	even $\rightarrow$ rational	2
2	3	$\sqrt{2^3}$	8	odd $\rightarrow$ irrational	$2\sqrt{2}$
3	4	$\sqrt{2^4}$	16	even $\rightarrow$ rational	4
3	5	$\sqrt{2^5}$	32	odd $\rightarrow$ irrational	$4\sqrt{2}$
4	6	$\sqrt{2^6}$	64	even $\rightarrow$ rational	8
4	7	$\sqrt{2^7}$	128	odd $\rightarrow$ irrational	$8\sqrt{2}$
5	8	$\sqrt{2^8}$	256	even $\rightarrow$ rational	16
5	9	$\sqrt{2^9}$	512	odd $\rightarrow$ irrational	$16\sqrt{2}$

Table 7: The unfolded chain. Five active halvings ($c = 1$ through $c = 5$) unfolding the nine active depths ($d = 1$ through $d = 9$), seated alongside the axiom at rest ($c = 0$, $d = 0$). The 2^d column shows the power of 2 sitting under each square root: $\sqrt{2^9} = \sqrt{512} = 16\sqrt{2}$, $\sqrt{2^8} = \sqrt{256} = 16$, down to $\sqrt{2^0} = \sqrt{1} = 1$ (the axiom). Even exponents carry rational readings; odd exponents carry irrational readings. Singular/dual is the direct consequence of how many depths each count owns.

8 Cumulative Mechanics: The Count-by-Count Inheritance Table

The chain is cumulative. Each count does not replace the partitions of prior counts — it inherits them and adds its own new layer on top. The singular count $c = 1$ performs the first halving and produces 2 partitions. Every dual count after it inherits every partition that came before, then performs its own halving to double the partitions again.

This cumulative view is the mechanical spine of the engine. It shows, at every count, (i) what the count inherits from prior counts, (ii) what the count adds of its own, (iii) how many depths the count owns, and (iv) the total depth-operations (parity readings) the count carries.

Reading the Table (Ct. Count)

- **Count** 1 **is the singular trigger.** One halving, one depth ($d = 1$, odd, irrational $\sqrt{2}$), exactly 2 partitions. That single act locks in dual operation for every later count.

- **Count** 2 **is the first full dual.** It inherits the 2 partitions from $c = 1$ and adds its own 4 partitions. Parity on the two new depths ($d = 2$ even, $d = 3$ odd) forces one rational and one irrational reading simultaneously.

- **Count** 3 **carries** 2 **and the** 4 **and its own** 8**.** The cumulative partition view is exact — count 3 sits on top of everything count 1 and count 2 produced, and performs its own halving to double the partitions again.

- **Count** 4 **carries** $2 + 4 + 8$ **and its own** 16**.** Same pattern. The inherited column is 14; the own column is 16; the total at this level is 16 (the new count's own layer), with everything beneath it still intact.

Ct. c	Type	Inherited from prior counts	Own new partitions / domains (2^c)	Total partitions at this level	Depths owned	Depth operations (parity readings)	Total "operations" in this count (cumulative view)
1	singular	—	2	2	1	1 (irrational)	2 (partitions) + 1 depth op
2	dual	from c=1 (2)	4	4	2	2 (1 rational + 1 irrational)	$2+4+2$ depth ops
3	dual	from c=1 (2) + c=2 (4)	8	8	2	2 (1 rational + 1 irrational)	$2+4+$ own $8+2$ depth ops
4	dual	from c=1–3 ($2+4+8=14$)	16	16	2	2 (1 rational + 1 irrational)	$2+4+8+$ own $16+2$ depth ops
5	dual (closure)	from c=1–4 ($2+4+8+16=30$)	32	32	2	2 (1 rational + 1 irrational)	$2+4+8+16+$ own $32+2$ depth ops

Table 8: The cumulative count-by-count inheritance table. Each count inherits all partitions from prior counts and adds its own new layer (2^c). Count 1 is the singular trigger: one halving, one depth ($d = 1$, odd $\to$ irrational $\sqrt{2}$), 2 partitions. That single act forces every later count into dual operation — one even/rational depth plus one odd/irrational depth at the same step — with no extra axiom needed.

- **Count 5 closes.** It carries $2 + 4 + 8 + 16 = 30$ from prior counts, adds its own 32, and produces the terminal pair $(16, 16\sqrt{2})$ with 32 domains.

Cumulative Partition Total Across the Active Chain

The total partitions performed across all five active counts is the sum of the own columns:

$$2 + 4 + 8 + 16 + 32 \;=\; 62 \;=\; 2^6 - 2.$$

This is a structural number of the first closure. The $2^6 = 64$ frame sits one count past the active chain; the active chain itself occupies the 62 partitions that live inside that frame, with the axiom seat's single unit at one end and the next-count boundary at the other. The number 64 recurs throughout the broader engine — it is the natural register that sits just past the first closure's active work, and every reading of it (cumulative partition sum, wall inventory, bit counts per quadrant) is the same 64 seen from a different direction.

Why the Mechanics Close

The engine is self-consistent because:

- the singular $c = 1$ seeds the dual behavior (parity on a single odd depth),
- the cumulative partitions give the domain multiplicity at every level (2^c own, $2^{c+1} - 2$ cumulative),
- the depth operations (parity applied to each owned depth) force the simultaneous rational/irrational reading at every dual count,

- and the $5 \times 9 = 45$ base mechanical unit (counts $\times$ active depths) closes both the square (512) and the sphere ($512\sqrt{2} \approx 724.077$) at the same terminal count.

The mirror chain of Part Six and the 4D inverse tensor of Part Seven are then inevitable. They are the same 1 read inward from the four terminal faces, using exactly the same cumulative mechanics shown here.

Part Two — The Two Operational Structures: Square and Sphere

Part One walked the chain count by count. Part Two reads the same chain as the two operational structures it produces. The structures are not built on top of the chain — they *are* the chain, split by parity:

- The **square structure** is the chain read at the even depths — the rational readings $1, 2, 4, 8, 16$.
- The **sphere structure** is the chain read at the odd depths — the irrational readings $\sqrt{2}, 2\sqrt{2}, 4\sqrt{2}, 8\sqrt{2}, 16\sqrt{2}$.

Both structures share the same c and d. Both close at count $c = 5$. The square closes at 512. The sphere closes at $512\sqrt{2} \approx 724.077$.

Master Table: The Chain Split by Parity

The chain is read here in the $\sqrt{2}$-coefficient view, which keeps $\sqrt{2}$ visible at every odd depth and lets the coefficient halve in front of it as d steps. Count 1 is singular. Counts 2 through 5 are dual — each carries one rational view (square) and one irrational view (sphere) simultaneously.

c	d	$\sqrt{2^d}$	$\sqrt{2}$**-coeff view**	**Parity → Reading**	**Square (rational)**	**Sphere (irrational)**
0	0	$\sqrt{2^0}$	1	even → rational (axiom)	1	—
1	1	$\sqrt{2^1}$	$\sqrt{2}$	odd → irrational	—	$\sqrt{2} \approx 1.414214$
2	2	$\sqrt{2^2}$	2	even → rational	2	—
2	3	$\sqrt{2^3}$	$2\sqrt{2}$	odd → irrational	—	$2\sqrt{2} \approx 2.828427$
3	4	$\sqrt{2^4}$	4	even → rational	4	—
3	5	$\sqrt{2^5}$	$4\sqrt{2}$	odd → irrational	—	$4\sqrt{2} \approx 5.656854$
4	6	$\sqrt{2^6}$	8	even → rational	8	—
4	7	$\sqrt{2^7}$	$8\sqrt{2}$	odd → irrational	—	$8\sqrt{2} \approx 11.313708$
5	8	$\sqrt{2^8}$	16	even → rational	16	—
5	9	$\sqrt{2^9}$	$16\sqrt{2}$	odd → irrational	—	$16\sqrt{2} \approx 22.627417$

Table 9: The full chain split by parity. The square column read top-to-bottom is the rational structure $1, 2, 4, 8, 16$. The sphere column read top-to-bottom is the irrational structure $\sqrt{2}, 2\sqrt{2}, 4\sqrt{2}, 8\sqrt{2}, 16\sqrt{2}$.

How the Chain Builds the Two Wholes

At each count c the chain has exactly 2^c domains: $1, 2, 4, 8, 16, 32$ as c advances from the axiom seat through the five active halvings. Every domain at that count carries one piece of the depth's value. The total at any count is therefore domains $\times$ piece — and at count $c = 5$ both totals close on the terminal wholes of the two structures.

Square Structure — Rationals Build the 512 Whole

c	d	**Piece** $\sqrt{2^d}$	**Domains** 2^c	**Domains** $\times$ **Piece**	**Total**
0	0	1	1	1×1	1
2	2	2	4	4×2	8
3	4	4	8	8×4	32
4	6	8	16	16×8	128
5	8	16	32	32×16	**512**

Table 10: The rational (square) build: each even depth's piece multiplied by the count's domain count. The terminal at $c = 5$, $d = 8$ closes the square at $32 \times 16 = 512$.

Sphere Structure — Irrationals Build the 724.077 Whole

c	d	**Piece** $\sqrt{2^d}$	**Decimal piece**	**Domains** 2^c	**Domains** $\times$ **Piece**	**Total**
1	1	$\sqrt{2}$	≈ 1.414214	2	$2 \times \sqrt{2}$	$2\sqrt{2} \approx 2.828427$
2	3	$2\sqrt{2}$	≈ 2.828427	4	$4 \times 2\sqrt{2}$	$8\sqrt{2} \approx 11.313708$
3	5	$4\sqrt{2}$	≈ 5.656854	8	$8 \times 4\sqrt{2}$	$32\sqrt{2} \approx 45.254834$
4	7	$8\sqrt{2}$	≈ 11.313708	16	$16 \times 8\sqrt{2}$	$128\sqrt{2} \approx 181.019336$
5	9	$16\sqrt{2}$	≈ 22.627417	32	$32 \times 16\sqrt{2}$	$\mathbf{512\sqrt{2} \approx 724.077}$

Table 11: The irrational (sphere) build: each odd depth's piece multiplied by the count's domain count. The terminal at $c = 5$, $d = 9$ closes the sphere at $32 \times 16\sqrt{2} = 512\sqrt{2} \approx 724.077$.

The Two Closures Together

At the terminal count $c = 5$ the chain delivers both structures simultaneously:

$$\underbrace{32 \times 16}_{\text{rational, } d=8} = \mathbf{512} \quad \text{(square)}, \qquad \underbrace{32 \times 16\sqrt{2}}_{\text{irrational, } d=9} = \mathbf{512\sqrt{2} \approx 724.077} \quad \text{(sphere)}.$$

The square is the chain's even-depth closure. The sphere is the chain's odd-depth closure. They share the same c and the same d ladder; they differ only in parity. Together they are the two operational structures of the first mechanical closure.

Why Count 5 / Depths 8–9 Is the First Closure

Count 5 is not chosen — it is forced. It is the first count at which everything in the engine lands together for the first time. Six things happen at once at $c = 5$, $d = 9$, and none of them happens any earlier:

1. **The nine active depths complete.** The five active halvings ($c = 1$ through $c = 5$) unfold exactly the nine active depths ($d = 1$ through $d = 9$). The axiom at $c = 0$, $d = 0$ sits at rest alongside them. The exponent ladder stops at $\sqrt{2^9}$, the first odd depth where the irrational coefficient reaches $16\sqrt{2}$.

2. **Terminal pair is clean and symmetric.** The pair $(16, 16\sqrt{2})$ is the first time the chain produces a rational and an irrational reading sharing the same coefficient 16, the same count $c = 5$, and the same domain count 32. One face on the square, one face on the sphere, perfectly paired.

3. **Both operational structures close simultaneously.** The square closes at $32 \times 16 = 512$ at depth $d = 8$; the sphere closes at $32 \times 16\sqrt{2} = 512\sqrt{2} \approx 724.077$ at depth $d = 9$. Neither structure closes at any earlier count.

4. **The integer 720 is machined.** The active partition (counts 1–5, depths 1–9) gives $5 \times 9 = 45$, and the binary ladder $45 \to 90 \to 180 \to 360 \to 720$ produces the integer 720 as a clean machined reading. This only appears as an integer at this boundary.

5. **Four parallel faces are ready.** The two operational structures, each with a machined side and an observed side, give the four faces (512, 514.90, 720, 724.077). All four become available at the same instant.

6. **The mirror engine can reflect.** With four terminal faces in hand and the constant per-unit slip -0.005663 established, the chain can now restart inward from any of the four faces (Part Six) and meet the outward chain back at the shared terminal pair $(16, 16\sqrt{2})$. The closed cycle is possible only at count 5.

Beyond the First Closure: Repetition, Not Novelty

The chain does not stop at count 5 in principle — c and d are unbounded, as established at the start of the paper. But nothing mechanically new appears past the first closure. Count 6 owns depths 10 and 11: still one even and one odd, still a dual count, still the same parity rule. The readings simply double: 32 and $32\sqrt{2}$, then 64 and $64\sqrt{2}$, then 128 and $128\sqrt{2}$. The domain count keeps doubling: $64, 128, 256, \ldots$. The same singular/dual behavior, the same mirror splitting, the same constant per-unit slip — everything repeats at larger scale.

The first closure is therefore the smallest complete engine. It contains:

- the axiom of 1 at rest,
- the full active chain ($c = 1$ through 5, $d = 1$ through 9): five halvings unfolding the nine active depths,
- both operational structures (square and sphere),
- the machined integer 720,

- the four terminal faces (512, 514.90, 720, 724.077),
- the constant per-unit slip $-0.005663 = 1 - \frac{32\sqrt{2}}{45}$,
- and the ability to mirror inward from the terminals back to the shared pair $(16, 16\sqrt{2})$.

Anything past count 5 runs the same engine at double the size. Because the pattern does not evolve — it scales — this paper works the first closure only. Higher closures are reserved for future papers.

Part Three — The Finished Chain

9 Summary Table

Count	Depth	Parity	Reading	Value	Domains
0	$\sqrt{2^0}$	even	axiom (unit at rest)	1	1
1	$\sqrt{2^1}$	odd	irrational singular	$\sqrt{2} \approx 1.414214$	2
2	$\sqrt{2^2}$	even	rational	2	4
2	$\sqrt{2^3}$	odd	irrational	$2\sqrt{2} \approx 2.828427$	4
3	$\sqrt{2^4}$	even	rational	4	8
3	$\sqrt{2^5}$	odd	irrational	$4\sqrt{2} \approx 5.656854$	8
4	$\sqrt{2^6}$	even	rational	8	16
4	$\sqrt{2^7}$	odd	irrational	$8\sqrt{2} \approx 11.313708$	16
5	$\sqrt{2^8}$	even	rational	16	32
5	$\sqrt{2^9}$	odd	irrational	$16\sqrt{2} \approx 22.627417$	32

Table 12: Five active halvings ($c = 1$ through $c = 5$) unfolding the nine active depths ($d = 1$ through $d = 9$), giving terminal exponent 9. The axiom seat at $c = 0$, $d = 0$ sits at rest alongside them. Count 1 owns depth 1 (the first irrational). Counts 2 through 5 each own two consecutive depths, one even and one odd. Active-depth tally: $1 + 2 + 2 + 2 + 2 = 9$.

Part Four — The 5×9 Active Partition Matrix and the Machining of 720

10 Active Partition

The axiom at $c = 0$, $d = 0$ is the starting condition of the chain, not part of the partition work the chain performs. The *active* partition consists of the counts that do the halving and the depths

that carry new readings:

- Active counts: $1, 2, 3, 4, 5$ — exactly 5 counts.
- Active depths: $\sqrt{2^1}$ through $\sqrt{2^9}$ — exactly 9 depths.

Count 1 is singular (one active depth: $\sqrt{2^1}$). Counts 2 through 5 are dual (each owns two active depths). The tally of active depths is

$$\underbrace{1}_{\text{count 1}} + \underbrace{2}_{\text{count 2}} + \underbrace{2}_{\text{count 3}} + \underbrace{2}_{\text{count 4}} + \underbrace{2}_{\text{count 5}} = 9.$$

11 The Cross-Product

The 5 active counts interact with the 9 active depths through a structural cross-product:

$$5 \text{ (active counts)} \times 9 \text{ (active depths)} = 45 \quad \text{base mechanical units.}$$

12 Binary Expansion Forcing 720

Because the chain is a sequence of partitions, every partition is a halving, and the mechanical base unit 45 sits on the binary doubling ladder:

$$\begin{array}{rl} \text{base unit} & 45 \\ \times 2 & 90 \\ \times 2 & 180 \\ \times 2 & 360 \\ \times 2 & \mathbf{720}. \end{array}$$

The integer 720 is machined from inside the chain:

$$\mathbf{5 \times 9 \times 16 = 720}.$$

Nothing is added from outside. The 5 active counts and the 9 active depths are the only two boundaries the chain has, and the binary ladder $\times 16$ does the rest.

13 Domain Multiplicity of the Machined 720

At each count the machined rational 720 is exactly tiled by a whole number of identical pieces. The number of pieces equals the number of domains at that count, which is 2^c where c is the count index.

Part Five — The Rational Face and the Irrational Face

The finished chain has two faces that are built simultaneously by the parity rule. The rational face is the machined integer 720. The irrational face is the terminal $512\sqrt{2} \approx 724.077$. Neither face is derived from the other; both are produced together by the same unfolding, one from the even depths and one from the odd depths.

Count	Type	Domains	Piece Size	Total
0	axiom (rest)	1	720	$1 \times 720 = 720$
1	singular	2	360	$2 \times 360 = 720$
2	dual	4	180	$4 \times 180 = 720$
3	dual	8	90	$8 \times 90 = 720$
4	dual	16	45	$16 \times 45 = 720$
5	dual	32	22.5	$32 \times 22.5 = 720$

Table 13: Domain multiplicity of the machined 720 across the five active halvings ($c = 1$ through $c = 5$), seated alongside the axiom at rest ($c = 0$). Domains at count c equal 2^c.

14 Side-by-Side Halving of the Two Faces

Halving each face uniformly at every count produces two columns that run in lock-step. The per-count slip between them is identical at every row.

Count	Rational (from 720)	Irrational (from $512\sqrt{2}$)	Differential	Per-unit slip
0	720.000000	724.077344	−4.077344	−0.005663
1	360.000000	362.038672	−2.038672	−0.005663
2	180.000000	181.019336	−1.019336	−0.005663
3	90.000000	90.509668	−0.509668	−0.005663
4	45.000000	45.254834	−0.254834	−0.005663
5	22.500000	22.627417	−0.127417	−0.005663

Table 14: The two faces of the finished chain, halved at each count. The differential column is the signed gap rational minus irrational; it halves in exact step with the two columns themselves. The per-unit slip (differential divided by rational) is constant at −0.005663 across all rows.

Every rational row has an irrational twin. The machined 720 sits underneath 724.077344. The quadrant 180 sits underneath 181.019336. The base unit 45 sits underneath 45.254834. And the half-unit 22.5 sits underneath 22.627417. Every rational unit in the ladder carries the same per-unit slip −0.005663. The gap is a property of the engine, not of any particular scale.

The Slip in Closed Form

The constant −0.005663 in the per-unit slip column is read two ways at once, and both readings give the same number. As a measurement it is what falls out when each face is halved in lockstep at every count: the gap divided by the rational reading is identical at every row. As a closed form it is the exact rational-to-irrational deviation from 1 written entirely in the chain's own coordinates:

$$-0.005663\ldots \;=\; 1 \;-\; \frac{32\sqrt{2}}{45}.$$

Every quantity in this expression is generated by the chain. The 32 is 2^c at the terminal count $c = 5$ — the chain's own terminal domain count. The 45 is the active count $\times$ active depth product 5×9 from Part Four — the base mechanical unit that machines 720. The $\sqrt{2}$ is the parity-forced irrational seed at $d = 1$. Nothing is imported. The slip is therefore both: a measured halving result and the chain's own internal deviation from 1. The two readings coincide because the engine is closed — the same number arrived at from two directions of reading is the engine confirming itself.

Part Six — The Mirror Chain: Restarting the Axiom at All Four Terminals

Parts One through Five read the inverse mechanical count *outward* from the axiom of 1. The same count can be read *inward* by restarting the axiom simultaneously at all four finished terminal faces: the rational terminal 512, the slipped rational ≈ 514.90, the machined rational 720, and the irrational terminal $512\sqrt{2} \approx 724.077$. The parity rule is preserved exactly on every face, and all four inward chains meet the outward chain at the shared terminal pair $(16, 16\sqrt{2})$ at count 5. The engine closes on itself across all four faces.

15 The Outside Axiom

When the axiom is restarted at any of the four terminal values, the axiom seat owns local depth $\sqrt{2^0}$ again. By the parity rule (even exponent) the reading is rational on the rational and slipped faces, and the corresponding scaled value appears on the machined and irrational faces. From that point forward the same parity rule drives the unfolding on every face. Count 1 owns local depth $\sqrt{2^1}$ (odd), producing the matching irrational reading on each face. Counts 2 through 5 each own two consecutive local depths and therefore carry one rational and one irrational reading simultaneously on every face.

16 Outside Unfolding Table

17 Conservation Check

At every count the appropriate terminal value multiplied by the number of domains is conserved on each face (512 on the rational terminal, ≈ 514.90 on the slipped face, 720 on the machined face, and $512\sqrt{2}$ on the irrational face).

18 Shared Terminal

The inside chain builds up from the axiom of 1 to the pair $(16, 16\sqrt{2})$ at count 5. The outside chains, started from any of the four terminal faces (512, ≈ 514.90, 720, ≈ 724.077), all contract down to the same shared terminal pair $(16, 16\sqrt{2})$ at count 5. All chains obey the same parity rule, unfold the same nine active depths, and close the engine perfectly across all four faces.

The four faces are the four parallel readings of the same chain (five active halvings unfolding the nine active depths, seated alongside the axiom at rest). The constant per-unit slip -0.005663 links them at every scale.

Count	Depth	Parity	Reading	Value (on 512 face)	Domains
0	$\sqrt{2^0}$	even	axiom (unit at rest)	512	1
1	$\sqrt{2^1}$	odd	irrational singular	$256\sqrt{2} \approx 362.039$	2
2	$\sqrt{2^2}$	even	rational	128	4
2	$\sqrt{2^3}$	odd	irrational	$128\sqrt{2} \approx 181.019$	4
3	$\sqrt{2^4}$	even	rational	64	8
3	$\sqrt{2^5}$	odd	irrational	$64\sqrt{2} \approx 90.510$	8
4	$\sqrt{2^6}$	even	rational	32	16
4	$\sqrt{2^7}$	odd	irrational	$32\sqrt{2} \approx 45.255$	16
5	$\sqrt{2^8}$	even	rational	16	32
5	$\sqrt{2^9}$	odd	irrational	$16\sqrt{2} \approx 22.627$	32

Table 15: Outside unfolding restarted simultaneously at all four terminal faces (512, $\approx$ 514.90, 720, $\approx$ 724.077). The parity rule and domain structure are preserved exactly on every face. The values shown are for the rational terminal 512 face; the other three faces scale proportionally while maintaining the same structure and constant per-unit slip.

Count	Domains	Rational conservation (512 face)	Irrational conservation ($512\sqrt{2}$ face)
0	1	$1 \times 512 = 512$	—
1	2	—	$2 \times 256\sqrt{2} = 512\sqrt{2}$
2	4	$4 \times 128 = 512$	$4 \times 128\sqrt{2} = 512\sqrt{2}$
3	8	$8 \times 64 = 512$	$8 \times 64\sqrt{2} = 512\sqrt{2}$
4	16	$16 \times 32 = 512$	$16 \times 32\sqrt{2} = 512\sqrt{2}$
5	32	$32 \times 16 = 512$	$32 \times 16\sqrt{2} = 512\sqrt{2}$

Table 16: Conservation on the rational and irrational faces at every count of the outside chain. The same conservation holds proportionally on the slipped and machined faces.

The engine is closed. Five active halvings unfolding nine active depths, one parity rule, one axiom of 1 at rest beneath them. Every other structure — the singular and dual counts, the terminal pair, the four parallel faces, the machined integer 720, the constant per-unit slip — is a consequence of unfolding the active-depth ladder 1 through 9 under the parity rule, starting from the axiom of 1.

Part Seven — Inverse Dual Hemisphere Quantum Mechanics: The 4D Inverse Tensor

19 The Two Operational Structures

Part Seven reads the finished chain as an **inverse 4D mechanical engine** with two operational structures running in parallel. Each structure has a machined side (exact rational closure) and an observed side (irrational twin carrying the chain's constant per-unit slip).

Structure	**Machined side**	**Observed side**	**Seam at axiom**
Inverse 4D mechanical **square**	512	$514.90\ldots$	$2.90\ldots$
Inverse 4D mechanical **sphere**	720	$724.077\ldots$	$4.077\ldots$

The machined side of the square is the exact rational closure $512 = 2^9$, the terminal 2^d value of the chain. Its observed twin $514.90\ldots$ is the same count structure read on the irrational face, carrying the constant per-unit slip -0.005663 row by row. The machined side of the sphere is the exact rational closure 720, machined from inside the chain by $5 \times 9 \times 16 = 720$ (Part Four). Its observed twin is $512\sqrt{2} \approx 724.077\ldots$, the terminal irrational reading of the chain.

The two sides are not independent number lines. They are the same inverse mechanical engine read on two geometries: the square structure reads the chain in bin counts, and the sphere structure reads it in degrees. Both structures obey the same parity rule, halve in the same lockstep across all five active halvings, and carry the same constant per-unit slip -0.005663. The sphere seam $724.077 - 720 = 4.077\ldots$ and the square seam $514.90 - 512 = 2.90\ldots$ are the same slip expressed on the two geometries; each halves in exact step with its own pair as the chain contracts.

Together, the machined pair (512, 720) and the observed pair (514.90, 724.077) form the **inverse 4D mechanical sphere and square** — the two operational structures of the finished chain. Every construction in the rest of Part Seven (the four indexed faces, the mirror-pair tables, the 4D tensor) is one of these two structures read at a specific count.

The Four Forms and the Unity Relation

The two operational structures give exactly four forms of the same single whole:

$$\underbrace{512}_{\text{machined square}}, \quad \underbrace{514.90\ldots}_{\text{observed square}}, \quad \underbrace{720}_{\text{machined sphere}}, \quad \underbrace{724.077\ldots}_{\text{observed sphere}}.$$

Each form is a complete reading of the axiom of 1. The machined side is the exact rational closure (the *observer*); the observed side is the irrational twin (the *observed*). Under the axiom of 1, the two sides of each structure are not two numbers — they are the same unity read on two geometries. This is the unity relation:

Unity Relation

$$\frac{\text{observed}}{\text{observer}} = 1$$

The square pair obeys it: $514.90/514.90 = 1$ and $512/512 = 1$, and the pair itself is one single engine. The sphere pair obeys it: $724.077/724.077 = 1$ and $720/720 = 1$, and the pair itself is one single engine. Across the two structures the same unity holds: each of the four forms is a complete 1, and every differential between them (the 2.90 square seam, the 4.077 sphere seam, and the constant per-unit slip -0.005663) is internal to that unity, not a departure from it. The differentials are the *shape* of the 1, not a break in it.

The finished chain possesses four parallel scalar faces: the rational terminal 512, the slipped rational ≈ 514.90, the machined rational 720, and the irrational terminal ≈ 724.077. These are not independent number lines. They are four signed real-number vectors that pivot around the single inverse center unit 0.

Every active halving after the axiom is dual: it splits every existing domain symmetrically across the new center unit 0, instantly creating a new mirror pair (decompression $\leftrightarrow$ compression). This is Inverse Dual Hemisphere Quantum Mechanics viewed as a 4D inverse tensor. The scalar walk tables of Part One give the magnitude face. This part gives the signed directional tensor face expressed in inverse unit values.

At each count c, the whole face value is divided into 2^c equal pieces. The axiom seat is the single unpartitioned whole. The geometric names and mirror-pair structure remain identical on every face; only the piece size changes. Conservation holds at every count on every face: domains $\times$ piece = the whole.

20 Rational Terminal Face: Machined Square 512

Machined Square

512

c	**Geometric name**	**Domains** 2^c	**Mirror pairs**	**Piece size**	**Domains $\times$ Piece**
0	axiom (single)	1	0	512	$1 \times 512 = 512$
1	dual (hemispheres)	2	1	256	$2 \times 256 = 512$
2	quadrant	4	2	128	$4 \times 128 = 512$
3	octant	8	4	64	$8 \times 64 = 512$
4	hexadecant	16	8	32	$16 \times 32 = 512$
5	dotriacontant (terminal)	32	16	16	$32 \times 16 = 512$

Table 17: Machined square 512 face. At every count the whole 512 is conserved: domains $\times$ piece = 512. The piece halves with each count; the domain count doubles.

21 Observed Square Face: 514.90

Observed Square

514.90 . . .

c	**Geometric name**	**Domains** 2^c	**Mirror pairs**	**Piece size**	**Domains** $\times$ **Piece**
0	axiom (single)	1	0	≈ 514.90	$1 \times 514.90 = 514.90$
1	dual (hemispheres)	2	1	≈ 257.45	$2 \times 257.45 = 514.90$
2	quadrant	4	2	≈ 128.725	$4 \times 128.725 = 514.90$
3	octant	8	4	≈ 64.3625	$8 \times 64.3625 = 514.90$
4	hexadecant	16	8	≈ 32.18125	$16 \times 32.18125 = 514.90$
5	dotriacontant (terminal)	32	16	≈ 16.090625	$32 \times 16.090625 = 514.90$

Table 18: Observed square ≈ 514.90 face. Conservation holds at every count: domains $\times$ piece ≈ 514.90.

22 Machined Sphere Face: 720

Machined Sphere

720

c	**Geometric name**	**Domains** 2^c	**Mirror pairs**	**Piece size**	**Domains** $\times$ **Piece**
0	axiom (single)	1	0	720	$1 \times 720 = 720$
1	dual (hemispheres)	2	1	360	$2 \times 360 = 720$
2	quadrant	4	2	180	$4 \times 180 = 720$
3	octant	8	4	90	$8 \times 90 = 720$
4	hexadecant	16	8	45	$16 \times 45 = 720$
5	dotriacontant (terminal)	32	16	22.5	$32 \times 22.5 = 720$

Table 19: Machined sphere 720 face. At every count the whole 720 is conserved: domains $\times$ piece $= 720$.

23 Observed Sphere Face: 724.077

Observed Sphere

$724.077\ldots$

24 Duality of the Mechanics

The axiom is single: the unbroken whole vector (the full face value itself). Every active count is dual: it partitions every existing domain, creating exactly as many new decompression $\leftrightarrow$ compression mirror pairs as there were domains in the previous count.

The number of mirror pairs therefore doubles at each step after the axiom ($0 \to 1 \to 2 \to 4 \to 8 \to 16$). This doubling is the direct mechanical signature of Inverse Dual Hemisphere Quantum Mechanics. The four faces differ only in piece size; the duality structure itself is identical.

c	**Geometric name**	**Domains** 2^c	**Mirror pairs**	**Piece size**	**Domains** $\times$ **Piece**
0	axiom (single)	1	0	≈ 724.077	$1 \times 724.077 = 724.077$
1	dual (hemispheres)	2	1	≈ 362.038672	$2 \times 362.039 = 724.077$
2	quadrant	4	2	≈ 181.019336	$4 \times 181.019 = 724.077$
3	octant	8	4	≈ 90.509668	$8 \times 90.510 = 724.077$
4	hexadecant	16	8	≈ 45.254834	$16 \times 45.255 = 724.077$
5	dotriacontant (terminal)	32	16	≈ 22.627417	$32 \times 22.627 = 724.077$

Table 20: Observed sphere ≈ 724.077 face. Conservation holds at every count: domains $\times$ piece ≈ 724.077.

The Inverse 4D Operating Tensor is now complete. It shows that the entire engine is a single signed directional structure driven by the same parity rule and the same dual-hemisphere mirror mechanics across all four faces.

25 Operation of the Mirror Pairs: Q1-Q3, Q2-Q4 and Higher

Every active count is dual. It does not split once — it splits **every existing domain** symmetrically across the inverse center unit 0. This creates perfect mirror pairs (decompression $\leftrightarrow$ compression).

The pattern is always the same: opposite domains mirror each other across the center. Q1 mirrors Q3, Q2 mirrors Q4, O1 mirrors O8, and so on.

The tables below show the concrete mirror pairs on the machined 720 face. The identical duality pattern appears on all four faces.

Axiom — Single

No split. Whole vector only.

Count 1 — Dual (hemispheres)

- Decompression hemisphere $\leftrightarrow$ Compression hemisphere: $-360 \leftrightarrow +360$

Count 2 — Quadrant

- Q1 $\leftrightarrow$ Q3: $+180 \leftrightarrow -180$
- Q2 $\leftrightarrow$ Q4: $+180 \leftrightarrow -180$

Count 3 — Octant

- O1 $\leftrightarrow$ O8: $+90 \leftrightarrow -90$
- O2 $\leftrightarrow$ O7: $+90 \leftrightarrow -90$
- O3 $\leftrightarrow$ O6: $+90 \leftrightarrow -90$
- O4 $\leftrightarrow$ O5: $+90 \leftrightarrow -90$

Count 4 — Hexadecant

- H1 $\leftrightarrow$ H16: $+45 \leftrightarrow -45$
- H2 $\leftrightarrow$ H15: $+45 \leftrightarrow -45$
- H3 $\leftrightarrow$ H14: $+45 \leftrightarrow -45$
- H4 $\leftrightarrow$ H13: $+45 \leftrightarrow -45$
- H5 $\leftrightarrow$ H12: $+45 \leftrightarrow -45$
- H6 $\leftrightarrow$ H11: $+45 \leftrightarrow -45$
- H7 $\leftrightarrow$ H10: $+45 \leftrightarrow -45$
- H8 $\leftrightarrow$ H9: $+45 \leftrightarrow -45$

Count 5 — Dotriacontant (terminal)

- D1 $\leftrightarrow$ D32: $+22.5 \leftrightarrow -22.5$
- D2 $\leftrightarrow$ D31: $+22.5 \leftrightarrow -22.5$
- D3 $\leftrightarrow$ D30: $+22.5 \leftrightarrow -22.5$
- D4 $\leftrightarrow$ D29: $+22.5 \leftrightarrow -22.5$
- D5 $\leftrightarrow$ D28: $+22.5 \leftrightarrow -22.5$
- D6 $\leftrightarrow$ D27: $+22.5 \leftrightarrow -22.5$
- D7 $\leftrightarrow$ D26: $+22.5 \leftrightarrow -22.5$
- D8 $\leftrightarrow$ D25: $+22.5 \leftrightarrow -22.5$
- D9 $\leftrightarrow$ D24: $+22.5 \leftrightarrow -22.5$
- D10 $\leftrightarrow$ D23: $+22.5 \leftrightarrow -22.5$
- D11 $\leftrightarrow$ D22: $+22.5 \leftrightarrow -22.5$
- D12 $\leftrightarrow$ D21: $+22.5 \leftrightarrow -22.5$
- D13 $\leftrightarrow$ D20: $+22.5 \leftrightarrow -22.5$
- D14 $\leftrightarrow$ D19: $+22.5 \leftrightarrow -22.5$
- D15 $\leftrightarrow$ D18: $+22.5 \leftrightarrow -22.5$
- D16 $\leftrightarrow$ D17: $+22.5 \leftrightarrow -22.5$

The mechanical rule never changes: each dual count doubles the domains and creates exactly as many new mirror pairs as there were domains before the split. Opposite domains always mirror each other across the inverse center unit 0. This is the direct cause of the 4D inverse tensor structure across all four faces.

26 The Binary Skeleton — Grouped Slash Notation for Counts and Depths

The unfolding of the ladder is driven by a single hierarchical binary engine. Both the *counts* and the *depths they own* are expressed in the same grouped slash notation. The axiom is the unit at rest and has no slot in the register — it is 1 with no partition performed. Count 1 introduces the first dual choice 0/1, unfolding the first active depth. Each subsequent dual count adds a fresh "00" block (reflecting the simultaneous ownership of one even and one odd depth). The terminal count 5 therefore carries exactly the grouped string

$$\mathbf{0/00/00/00/00}.$$

This string owns the final pair of depths (8 and 9) in the nine-slot active-depth register

$$0/0/0/0/0/0/0/0/0,$$

where each slot corresponds to one active depth $d = 1, 2, \ldots, 9$, and the rightmost bit of any slot is the parity bit forced by the parity rule.

The following table presents the complete binary skeleton in this native grouped notation. Every entry is a direct mechanical consequence of the axiom of 1 and the single parity rule (final bit = 0 for rational, 1 for irrational).

Count	**Count bin.**	**Domains**	**Depth**	**Depth binary (9 slots)**	**Parity**	**Reading**
0	axiom (rest)	1	—	— (no slot, unit at rest)	even (0)	1
1	0/1	2	1	0/0/0/0/0/0/0/0/1	odd (1)	$\sqrt{2}$
2	00	4	2	0/0/0/0/0/0/0/1/0	even (0)	2
			3	0/0/0/0/0/0/0/1/1	odd (1)	$2\sqrt{2}$
3	0/00	8	4	0/0/0/0/0/1/0/0/0	even (0)	4
			5	0/0/0/0/0/1/0/0/1	odd (1)	$4\sqrt{2}$
4	0/00/00	16	6	0/0/0/0/1/0/0/0/0	even (0)	8
			7	0/0/0/0/1/0/0/0/1	odd (1)	$8\sqrt{2}$
5	**0/00/00/00/00**	32	8	0/0/0/1/0/0/0/0/0	even (0)	16
			9	0/0/0/1/0/0/0/0/1	odd (1)	$16\sqrt{2}$

Table 21: The binary skeleton of the chain expressed in grouped slash notation. Counts and depths share the identical hierarchical structure. Count 5 carries exactly 0/00/00/00/00 and selects the final two slots of the nine-slot active-depth register. The axiom is the unit at rest and does not occupy a slot. The rightmost bit of each depth string is forced by the parity rule and determines the rational/irrational reading.

This grouped view makes three facts immediate. First, the domain growth $1 \to 2 \to 4 \to 8 \to 16 \to 32$ and the nine-slot active-depth register are two parallel binary ladders locked together by the same slash blocks. Second, the parity rule is literally the final bit of the grouped depth string (0 = rational, 1 = irrational). Third, the entire structure — singular/dual counts, the terminal pair $(16, 16\sqrt{2})$, the machining of 720, the two operational structures, the four parallel faces, and the 4D inverse tensor of Part Seven — is a necessary consequence of unfolding the nine-slot active-depth register under the single parity rule. No additional axioms are required.

The same grouped binary skeleton reappears unchanged in the outside mirror chain (Part Six) and in the inverse-unit indexing of the four terminal faces. It is the mechanical heartbeat shared by every part of the engine.